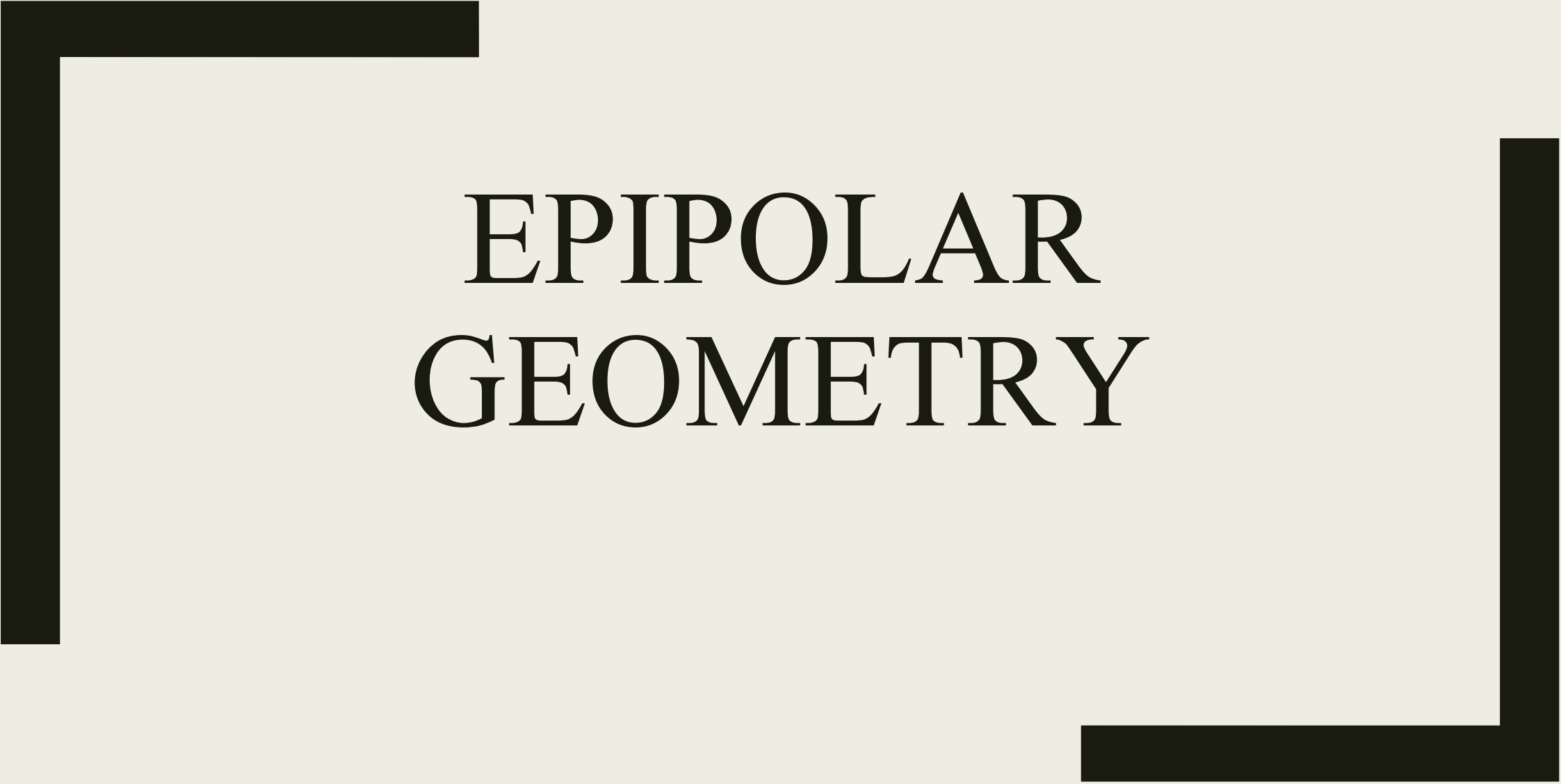
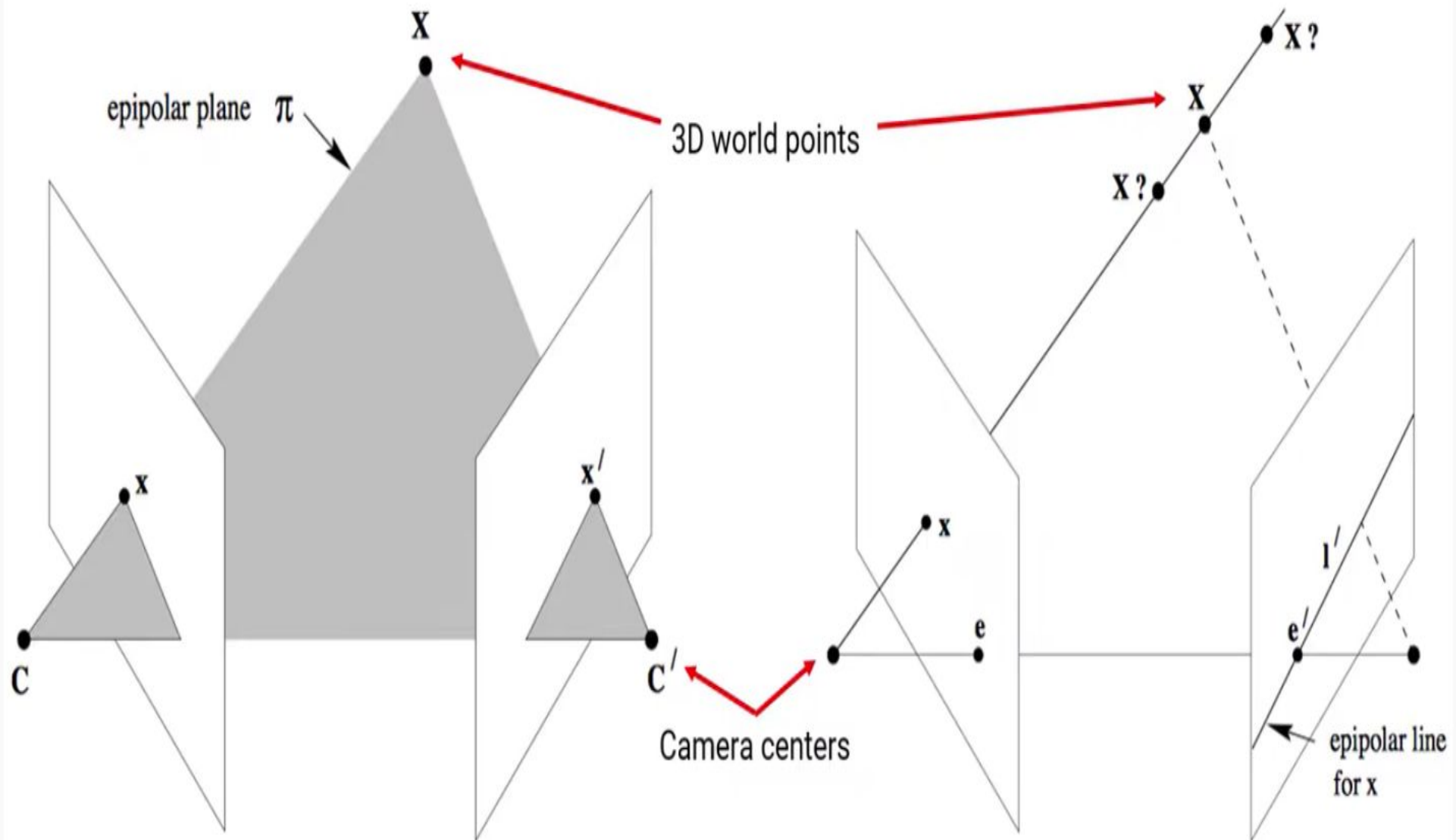


A thick black L-shaped frame is positioned on the left and bottom edges of the slide, framing the central text.

EPIPOLAR GEOMETRY

Introduction

- **Epipolar Geometry** as the mathematical bridge between two eyes. When you look at an object from two different angles, there is a strict geometric relationship between where a point appears in the left image and where it *must* appear in the right image.
- It is the fundamental logic used in 3D reconstruction, stereo vision (like in self-driving cars), and motion tracking.



Key Components

- **Baseline:** The straight line connecting the two camera centers (C_1 and C_2).
- **Epipoles (e_1, e_2):** These are the points where the baseline intersects the image planes. Essentially, it is the projection of one camera's center onto the other camera's image.
- **Epipolar Plane:** The triangle-shaped plane formed by the 3D point in the world (X) and the two camera centers.
- **Epipolar Line:** The intersection of the epipolar plane with the image plane.

Consider:

Two camera centers: C and C'

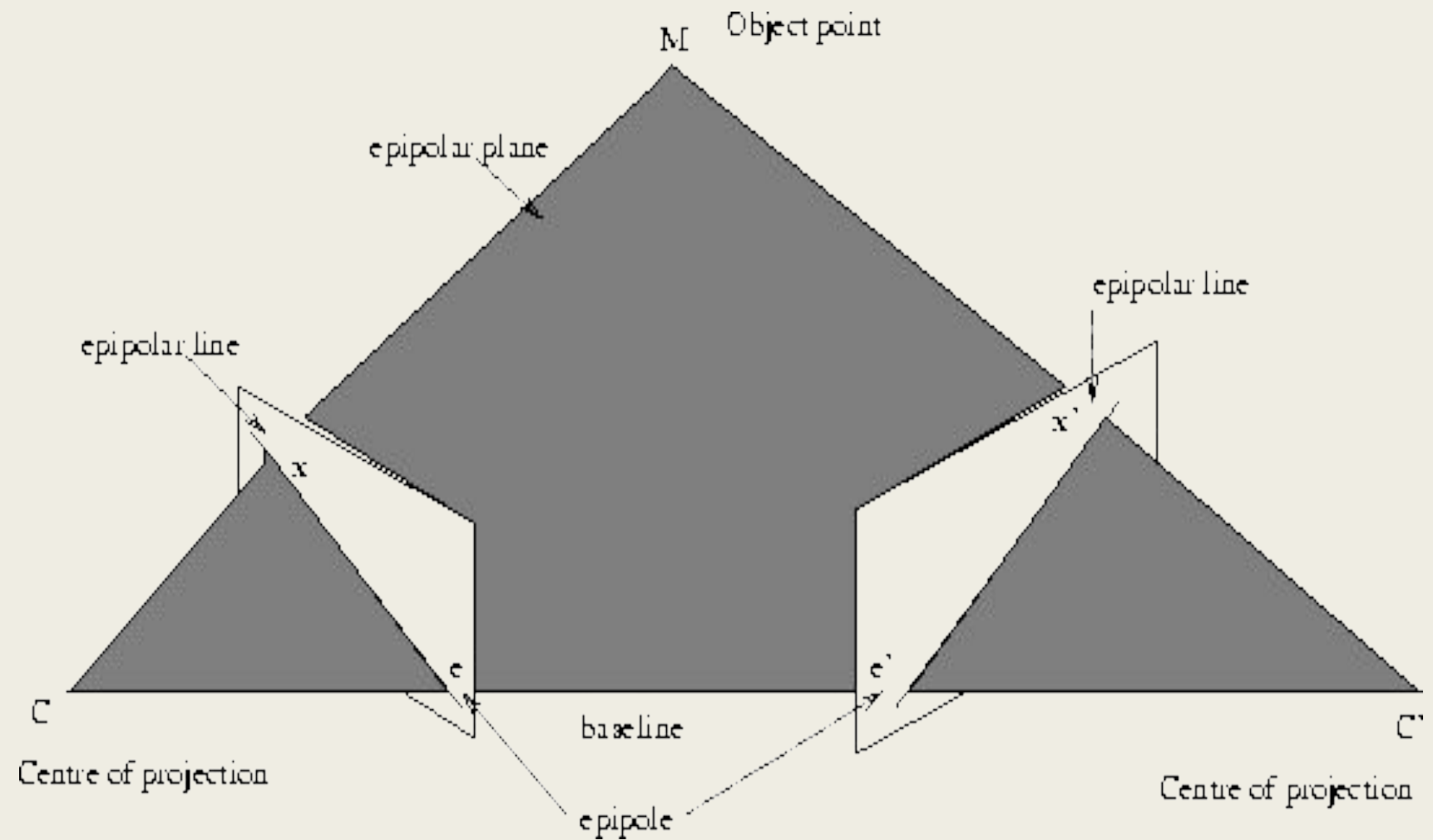
A 3D point M

Image projections:

x in image 1

x' in image 2

The key geometric structures:



- Epipolar Plane

The plane formed by: C_1 , C_2 , X

- Epipolar Lines

The intersection of the epipolar plane with each image plane.

If a point x is detected in image 1, its match in image 2 must lie on the corresponding epipolar line.

- Epipoles

$$x'^T F x = 0$$

Fundamental matrix

$$\mathbf{x}_i'^T \mathbf{F} \mathbf{x}_i = 0$$

Epipolar constraint
or correspondence condition

$i = 1, 2, \dots, m$

$$[x_i' \ y_i' \ 1] \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = 0$$

3 × 3 matrix
(9 unknowns)

$$x_i x_i' f_{11} + x_i y_i' f_{21} + x_i f_{31} + y_i x_i' f_{12} + y_i y_i' f_{22} + y_i' f_{23} + x_i' f_{31} + y_i f_{32} + f_{33} = 0$$

m correspondences

$$x_i x_i' f_{11} + x_i y_i' f_{21} + x_i f_{31} + y_i x_i' f_{12} + y_i y_i' f_{22} + y_i' f_{23} + x_i' f_{31} + y_i f_{32} + f_{33} = 0$$

For m correspondences

For m correspondences

$$A \begin{bmatrix} x_1 x'_1 & x_1 y'_1 & x_1 & y_1 x'_1 & y_1 y'_1 & y_1 & x'_1 & y'_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_m x'_m & x_m y'_m & x_m & y_m x'_m & y_m y'_m & y_m & x'_m & y'_m & 1 \end{bmatrix} \begin{bmatrix} f_{11} \\ f_{21} \\ f_{31} \\ f_{12} \\ f_{22} \\ f_{32} \\ f_{13} \\ f_{23} \\ f_{33} \end{bmatrix} = 0$$

Essential Matrix

- 3 x 3 matrix with additional property
- Obeys Pin hole camera model
- F – original image space
- E – normalized image coordinate
- Relative camera poses between two views are computed using E matrix
- $E = K^T F K$
- $E = K'^T F K$